

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Find out about the tornadoes that wreaked havoc, investigate the shape of the universe, unravel the methane mystery on mars and find out if eating red meat is good for you.



Smelling out aliens

Phosphine is a stinky and toxic gas that is found in dung heaps, swamps and bogs. This flammable gas is a waste product of life forms that do not breathe oxygen. Phosphine could thus be a promising biosignature to find aliens, as they can only be produced by these anaerobic organisms.

<https://dgit.in/alienstink>

WHAT'S NEW

Stanford engineers rig autonomous DeLorean to improve rider safety

Engineers from Stanford have been teaching an autonomous DeLorean manoeuvres that only the most accomplished human drivers can manage, with the goal of improving safety. This was done on a kilometre long obstacle course, made up of traffic cones. The DeLorean was trained to drift smoothly through the curves in the course, using only GPS coordinates and an algorithm in a laptop. There are two GPS antennas on board, that provide a geolocation accuracy with a resolution of one inch. The brakes are electronically controlled for greater precision, and the suspen-

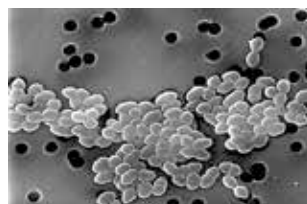


sion has been upgraded to handle the complex manoeuvres. Incidentally, the

DeLorean is named Marty. Marty can make donuts with mathematical precision. This is well beyond the expectations from most autonomous cars, which are programmed to do relatively simple tasks such as staying on a lane, avoiding obstacles, and maintaining certain distances from other vehicles. <https://dgit.in/stanforddrift>

Drug resistant superbug dispatches scouts to warn of an imminent attack

Vancomycin-resistant Enterococci or VRE is a superbug that developed resistance to a new antibiotic within a short span of five years. After spending over eight years studying the bacteria, the researchers found something surprising. The breakthrough came when the cells of VRE were grown in a flask, then killed off. When new cells were



introduced to the flask, they gained the resistance too. This showed that there was something in the medium that conferred the resistance. The

bacteria sends out sentinel proteins into the environment, that detects the drugs, as well as peptides produced by human macrophages. The sentinel warns of attack from drugs, but can also detect attacks from the natural immune system. The sentinel protein then warn the cells of the bugs of the imminent attack.

<https://dgit.in/superbugscout>



Bennu is active

NASA's OSIRIS-REX spacecraft has discovered that Bennu is an active after spending only weeks in orbit. The asteroid tends to periodically eject particles of dust, and dozens of these events have been observed. This is the first time that such activity has been recorded so precisely.

<https://dgit.in/benact>



Stable polymeric nitrogen

Researchers at Drexel have managed to create and stabilise pure polymeric nitrogen, which were so far only theorised to exist. The atomic bonds within nitrogen molecules could be a source of clean energy, with only harmless nitrogen as a byproduct.

<https://dgit.in/polynitro>



Cooler Lego in the universe

Researchers put in a bunch of Lego bricks into the dilution refrigerator at Lancaster University, and reduced the blocks to a shade above absolute zero. The thermal properties of the Lego bricks could prove to be useful in the building of future quantum computers.

<https://dgit.in/coolego>